INF398 - Introducción al aprendizaje automático

LDA/QDA

Alejandro Veloz

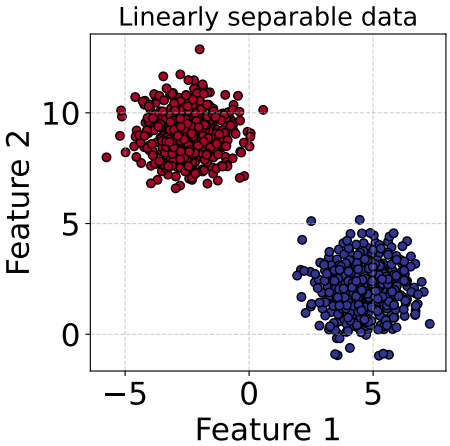
LDA / QDA

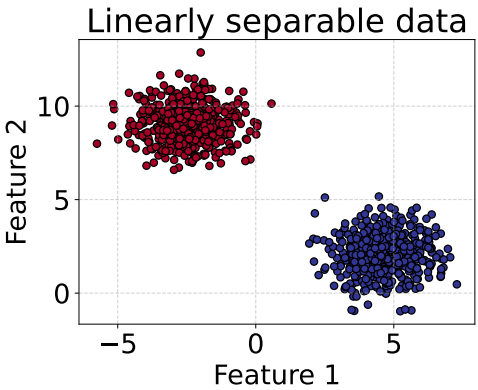
Two classic methods for classification based on the Bayesian approach:

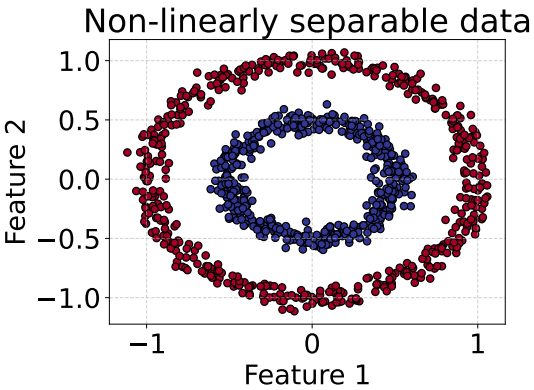
- Linear Discriminant Analysis (LDA).
- Quadratic Discriminant Analysis (QDA).

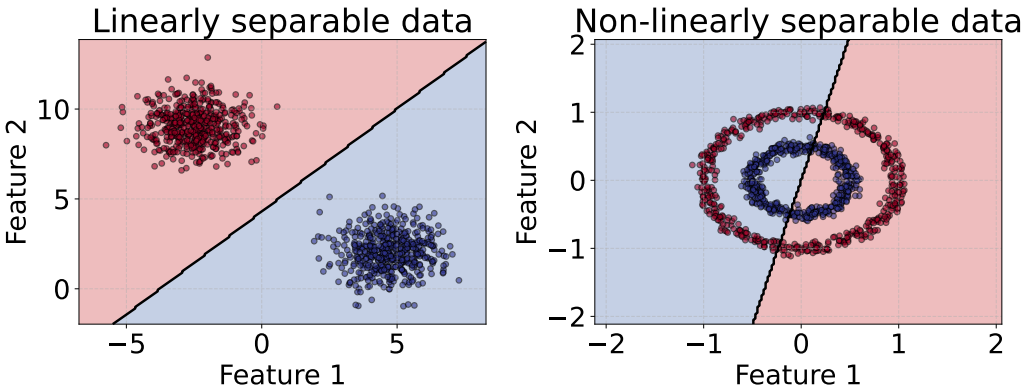
Linear discriminant analysis

- A linear classifier that separates data into different categories or classes denoted by $i = 1, 2, \ldots, K$.
- It is assumed that the data are linearly separable.
- LDA builds linear separation or decision boundaries (a linear combination of the features).




Linearly separable data
Feature 2
Feature 1
10
5
0
−5
0
5


Non-linearly separable data
Feature 2
Feature 1
1.0
0.5
0.0
−0.5
−1.0
−1
0
1


Linearly separable data
Feature 2
Feature 1
10
5
0
−5
0
5
Non-linearly separable data
Feature 2
Feature 1
2
1
0
−1
−2
−2
−1
0
1
2

Linear discriminant analysis

The method proposes assigning an object $\mathbf{x}$ to the group i that yields the highest conditional probability.

If there are g groups, Bayes' rule dictates that $\mathbf{x}$ is assigned to group i if $P(i \mid \mathbf{x}) > P(j \mid \mathbf{x})$ for all other groups $j \neq i$.

To implement this rule, the posterior probability $\mathrm{P}(i \mid \mathbf{x})$ must be known, which is typically difficult to obtain in practice from a set of observations of $\{\mathbf{x}_i, y_i\}_{i=1}^{N}$.

LDA: discriminant function

If we have K groups, Bayes' rule minimizes the total classification error by assigning the object $\mathbf{x}$ to the class i that has the highest conditional probability, i.e., $\mathrm{P}(i \mid \mathbf{x}) > \mathrm{P}(j \mid \mathbf{x}),\ \forall j \neq i$.

Since we do not have $\mathrm{P}(i \mid \mathbf{x})$, we can use Bayes' theorem, which states:

$$P(i \mid \mathbf{x}) = \frac{P(\mathbf{x} \mid i)P(i)}{\sum_{\forall j} P(\mathbf{x} \mid j)P(j)}$$

Therefore, we have that $\mathbf{x}$ is assigned to class i if:

$$\frac{P(\mathbf{x} \mid i)P(i)}{\sum_{\forall k} P(\mathbf{x} \mid k)P(k)} > \frac{P(\mathbf{x} \mid j)P(j)}{\sum_{\forall k} P(\mathbf{x} \mid k)P(k)},\ \forall j \neq i$$

$$P(\mathbf{x} \mid i)P(i) > P(\mathbf{x} \mid j)P(j),\ \forall j \neq i$$

LDA: discriminant function

Our assumption will be that the data comes from a multivariate Normal distribution.

$$P(\mathbf{x} \mid i) = \frac{1}{(2\pi)^{\frac{p}{2}} \left|\Sigma_i\right|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}\left(\mathbf{x}-\mu_i\right)^\top \Sigma_i^{-1}\left(\mathbf{x}-\mu_i\right)\right)$$

Replacing this distribution in Bayes' rule, we have:

$$\frac{P(i)}{(2\pi)^{\frac{p}{2}} \left|\Sigma_i\right|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}\left(\mathbf{x}-\mu_i\right)^\top \Sigma_i^{-1}\left(\mathbf{x}-\mu_i\right)\right) > \frac{P(j)}{(2\pi)^{\frac{p}{2}} \left|\Sigma_i\right|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}\left(\mathbf{x}-\mu_j\right)^\top \Sigma_j^{-1}\left(\mathbf{x}-\mu_j\right)\right), \forall j \neq i$$

LDA: discriminant function

$$\frac{P(i)}{|\Sigma_i|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}\left(\mathbf{x}-\mu_i\right)^\top \Sigma_i^{-1}\left(\mathbf{x}-\mu_i\right)\right) >$$

$$\frac{P(j)}{|\Sigma_j|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}\left(\mathbf{x}-\mu_j\right)^\top \Sigma_j^{-1}\left(\mathbf{x}-\mu_j\right)\right), \ \forall j \neq i$$

$$-\frac{1}{2}\ln\left(|\Sigma_i|\right) + \ln(P(i)) - \frac{1}{2}\left(\mathbf{x}-\mu_i\right)^\top \Sigma_i^{-1}\left(\mathbf{x}-\mu_i\right) >$$

$$-\frac{1}{2}\ln\left(|\Sigma_j|\right) + \ln(P(j)) - \frac{1}{2}\left(\mathbf{x}-\mu_j\right)^\top \Sigma_j^{-1}\left(\mathbf{x}-\mu_j\right), \ \forall j \neq i$$

LDA: discriminant function

$$\ln\left(|\Sigma_i|\right) - 2\ln(P(i)) + (\mathbf{x} - \mu_i)^\top \Sigma_i^{-1} (\mathbf{x} - \mu_i) <$$
$$\ln\left(\left|\Sigma_j\right|\right) - 2\ln(P(j)) + \left(\mathbf{x} - \mu_j\right)^\top \Sigma_j^{-1} \left(\mathbf{x} - \mu_j\right), \ \forall j \neq i.$$

LDA: discriminant function

Let,

$$d_i(\mathbf{x}) = \ln\left(|\Sigma_i|\right) + \left(\mathbf{x} - \mu_i\right)^\top \Sigma_i^{-1} \left(\mathbf{x} - \mu_i\right).$$

Then, the discriminant function becomes:

$$d_i(\mathbf{x}) - 2\ln(P(i)) < d_j(\mathbf{x}) - 2\ln(P(j)), \ \forall i \neq j.$$

If we assume equal covariance matrices $\Sigma = \Sigma_i = \Sigma_j$, then the discriminant function can be simplified:

$$\ln(|\Sigma|) - 2\ln(P(i)) + \left(\mathbf{x} - \mu_i\right)^\top \Sigma^{-1} \left(\mathbf{x} - \mu_i\right) <$$
$$\ln(|\Sigma|) - 2\ln(P(j)) + \left(\mathbf{x} - \mu_j\right)^\top \Sigma^{-1} \left(\mathbf{x} - \mu_j\right), \ \forall i \neq j.$$

$$\ln(|\Sigma|) - 2\ln(P(i)) + (\mathbf{x} - \mu_i)^\top \Sigma^{-1} (\mathbf{x} - \mu_i) <$$
$$\ln(|\Sigma|) - 2\ln(P(j)) + \left(\mathbf{x} - \mu_j\right)^\top \Sigma^{-1} \left(\mathbf{x} - \mu_j\right), \ \forall i \neq j.$$

Given that,

$$(\mathbf{x} - \mu_i)^\top \Sigma^{-1} (\mathbf{x} - \mu_i) = \mathbf{x}\Sigma^{-1}\mathbf{x}^\top - 2\mu_i\Sigma^{-1}\mathbf{x}^\top + \mu_i\Sigma^{-1}\mu_i^\top,$$

we have:

$$\ln(|\Sigma|) - 2\ln(P(i)) + \mathbf{x}\Sigma^{-1}\mathbf{x}^\top - 2\mu_i\Sigma^{-1}\mathbf{x}^\top + \mu_i\Sigma^{-1}\mu_i^\top <$$
$$\ln(|\Sigma|) - 2\ln(P(j)) + \mathbf{x}\Sigma^{-1}\mathbf{x}^\top - 2\mu_i\Sigma^{-1}\mathbf{x}^\top + \mu_i\Sigma^{-1}\mu_i^\top, \ \forall i \neq j.$$

The terms $\ln(|\Sigma|)$ and $\mathbf{x}\Sigma^{-1}\mathbf{x}^\top$ do not contribute to the decision of the discriminant function and are eliminated from the previous expression, therefore:

$$
\begin{aligned}
&-2\ln(P(i)) - 2\mu_i\Sigma^{-1}\mathbf{x}^\top + \mu_i\Sigma^{-1}\mu_i^\top < \\
&\qquad -2\ln(P(j)) - 2\mu_i\Sigma^{-1}\mathbf{x}^\top + \mu_i\Sigma^{-1}\mu_i^\top, \ \forall i \neq j, \\
&\ln(P(i)) + \mu_i\Sigma^{-1}\mathbf{x}^\top - \frac{1}{2}\mu_i\Sigma^{-1}\mu_i^\top > \\
&\qquad \ln(P(j)) + \mu_i\Sigma^{-1}\mathbf{x}^\top - \frac{1}{2}\mu_i\Sigma^{-1}\mu_i^\top, \ \forall i \neq j.
\end{aligned}
$$

LDA: discriminant function

Consequently, the discriminant function for class i is:

$$f_i = \mu_i \Sigma^{-1} \mathbf{x}^\top - \frac{1}{2} \mu_i \Sigma^{-1} \mu_i^\top + \ln(P(i)).$$

The object $\mathbf{x}$ is assigned to class i, if:

$$f_i > f_j, \ \forall i \neq j.$$

Under the assumption that the data comes from a multivariate Normal distribution and that all classes share the same covariance matrix.

LDA algorithm

1. Let $S = \{\mathbf{x}_i, y_i\}_{i=1}^N$ be a set of N observations.
2. Let N_k be the number of observations belonging to the k-th class.
3. Estimate the prior probability for the k-th class:

$$\hat{P}_k = \frac{N_k}{N}.$$

4. Estimate the mean vector of each class: $\hat{\mu}_k = \frac{1}{N_k} \sum_{y_i = k} \mathbf{x}_i$.
5. Estimate the covariance matrix:

$$\hat{\Sigma} = \frac{1}{N-K} \sum_k \sum_{G_i = k} (\mathbf{x}_i - \hat{\mu}_k)(\mathbf{x}_i - \hat{\mu}_k)^\top.$$

LDA: algorithm

To classify a (new) data point $\mathbf{x}$.

FOR $k = 1, \dots, K$ DO:

- Evaluate the discriminant function:

$$f_k = -\frac{1}{2}\left(\mathbf{x} - \hat{\mu}_k\right)^\top \hat{\Sigma}^{-1}\left(\mathbf{x} - \hat{\mu}_k\right) + \ln(P(k)).$$

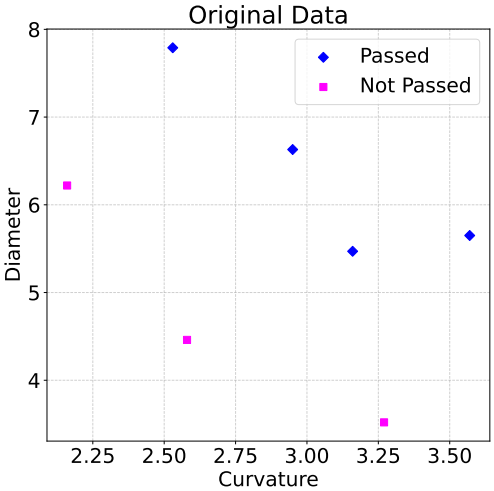
- Return: $G(\mathbf{x}) = \arg\max_k f_k$

END FOR

Example

Curvature	Diameter	Quallity Control Result
2,95	6,63	Passed
2,53	7,79	Passed
3,57	5,65	Passed
3,16	5,47	Passed
2,58	4,46	Not Passed
2,16	6,22	Not Passed
3,27	3,52	Not Passed

Example


Original Data
Passed
Not Passed
Diameter
Curvature
8
7
6
5
4
2.25
2.50
2.75
3.00
3.25
3.50

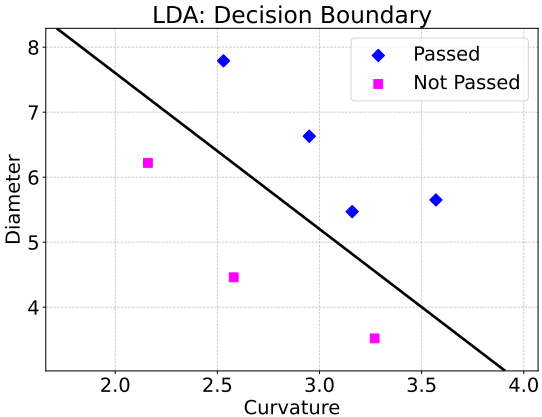
$$\hat{\mu}_1 = \begin{pmatrix} 3,05 \\ 6,39 \end{pmatrix} \quad \hat{\mu}_2 = \begin{pmatrix} 2,67 \\ 4,73 \end{pmatrix}$$

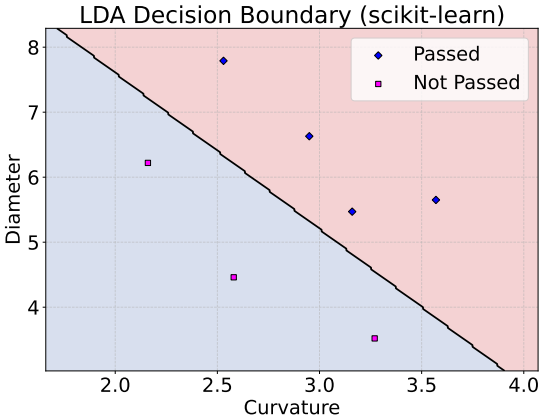
$$\hat{\Sigma}_1 = \begin{pmatrix} 0,56 & -1,24 \\ -1,24 & 3,41 \end{pmatrix} \quad \hat{\Sigma}_2 = \begin{pmatrix} 3,76 & -1,46 \\ -1,46 & 3,76 \end{pmatrix}$$

$$\hat{\Sigma} = \frac{1}{5}\left(\hat{\Sigma}_1 + \hat{\Sigma}_2\right)$$

Función discriminante

Curvature	Diameter	Class	f_1 (Passed)	f_2 (Not Passed)
2,95	6,63	Passed	-0,58	-16,27
2,53	7,79	Passed	-1,25	-18,75
3,57	5,65	Passed	-1,58	-23,35
3,16	5,47	Passed	-1,66	-9,48
2,58	4,46	Not Passed	-22,40	-1,41
2,16	6,22	Not Passed	-13,63	-1,65
3,27	3,52	Not Passed	-14,05	-1,65


LDA: Decision Boundary
Passed
Not Passed
Diameter
Curvature
8
7
6
5
4
2.0
2.5
3.0
3.5
4.0


LDA Decision Boundary (scikit-learn)
Passed
Not Passed
Diameter
Curvature
8
7
6
5
4
2.0
2.5
3.0
3.5
4.0

Class	Discriminant Function f_i(x)
Passed	156.383*Curv + 63.345*Diam + (-441.467)
Not Passed	127.597*Curv + 51.352*Diam + (-292.723)

QDA

The QDA technique maintains the assumption of Gaussianity for $P(\mathbf{x} \mid i)$.

It removes the assumption that every class has the same covariance matrix.

This means that $\Sigma_i \neq \Sigma_j, \ \forall i \neq j$.

Consequently, the discriminant function is defined as:

$$f_i(\mathbf{x}) = -\frac{1}{2}\ln(|\Sigma_i|) + \ln(P(i)) - \frac{1}{2}(\mathbf{x} - \mu_i)^\top \Sigma_i^{-1}(\mathbf{x} - \mu_i).$$

In this case, the discriminant functions are (hyper-) quadratic in $\mathbf{x}$; therefore, a higher separation power is generated compared to the linear case.

QDA algorithm

1. Let $S = \{\mathbf{x}_i, y_i\}_{i=1}^N$ be a set of N observations.
2. Let N_k be the number of observations belonging to the k-th class.
3. Estimate the prior probability for the k-th class:

$$\hat{P}_k = \frac{N_k}{N}.$$

4. Estimate the mean vector for the k-th class:

$$\hat{\mu}_k = \frac{1}{N_k} \sum_{G_i = k} \mathbf{x}_i.$$

5. Estimate the covariance matrix for the k-th class:

$$\hat{\Sigma}_k = \frac{1}{N_k - 1} \sum_{G_i = k} (\mathbf{x}_i - \hat{\mu}_k)(\mathbf{x}_i - \hat{\mu}_k)^\top.$$

QDA algorithm

To classify a (new) point $\mathbf{x}$:

FOR $k = 1, \dots, K$ DO:

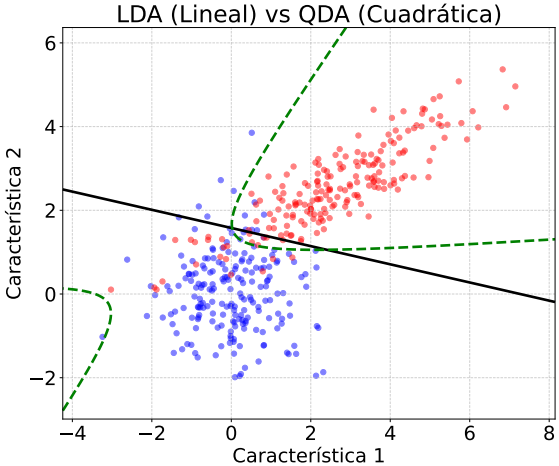
- Evaluate the discriminant function:

$$f_k(\mathbf{x}) = -\frac{1}{2}\ln(|\hat{\Sigma}_k|) + \ln(P(k)) - \frac{1}{2}(\mathbf{x} - \hat{\mu}_k)^\top \hat{\Sigma}_k^{-1}(\mathbf{x} - \hat{\mu}_k)$$

- RETURN:

$$G(\mathbf{x}) = \arg\max_k f_k.$$

END FOR


LDA (Lineal) vs QDA (Cuadrática)
Característica 2
Característica 1
6
4
2
0
−2
−4
−2
0
2
4
6
8